AI ACADEMY · LEVEL 2 · GRADE 7 · WEEK 5 · TEACHER GUIDE

Loops and Stopping

Learn to understand, design, build, test and explain AI systems responsibly.

Project: Transparent Book Recommender

THE CONNECTED WEEK

Teach while-loop stopping as a reasoned state change.

Use a complete decreasing-count example and finite defect prefixes, then assess the separate increasing loop.

Keep one habit: Look for pre-update checks, boundary crossing and a justified stopping argument.

MAIN LESSON · SECTION 1

Prepare this connected week

> Use one main plan and keep practice separate from the independent first attempt.

Learning goal and precision

Primary target: Trace variable values before and after each **loop iteration**

Make while-**loop** stopping the main target. Use the supplied decreasing count with a pre-update condition, a visible progress argument and zero/one/several-update tests. Treat the earlier range/accumulator material as labelled review. Never execute an intentionally

unbounded defect; inspect a finite paper prefix. Collect the selected increasing-**loop** case independently before feedback.

By the end, I can trace a while-**loop** and recognise when its **update** cannot satisfy the stopping test.

What to prepare

- Use the supplied scenario, student illustrations and blank paper. For numeric work, offer counters or a calculator after learners show the setup. Fictional records are supplied; personal data and paid accounts are unnecessary.
- Case exposure check: this source example appears in ai-2/4, ai-2/5, python-bridge/6. Reuse a familiar case for recall or modelling; verify that the case used to assess transfer has not already received feedback.
- Python 3 editor and **loop**s.py
- Counters or numbered paper cards
- Blank **trace table** with turn total before and total after columns

Main teaching plan

Session · Sequence and minutes · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Sequence and minutes:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

> **Sequence and minutes:** 0–5: retrieve the procedure; 5–15: W2 completion practice; 15–23: explain and check with a partner; 23–30: each learner answers the misconception check; 30–40: targeted reteaching or a harder variation; 40–45: prepare the investigation.
>
> **Resources:** Use Find and repair a mistake, Practise together, Run the investigation and Build your understanding; use the matching workbook W2 and misconception check.

> **Session:** Session 3 · Investigate and transfer · 45 min
>
> **Sequence and minutes:** 0–5: retrieve the decision rule; 5–20: investigate or trace; 20–30: start the independent case; 30–38: connect the project; 38–43: act on feedback; 43–45: plan the return check.
>
> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

This is a starting allocation of 135 minutes, not a validated duration. Preserve all named topics. Schedule extra time for connected examples, the full investigation, independent cases and project building.

Repeated copies of a case share one recorded attempt; they are not new **evidence**.

Opening retrieval and readiness

Without opening the old lesson, explain one key idea from **Trace Table**s. Give an example and a mistake that the idea helps you avoid.

Look for: Trace variable values before and after each **loop iteration**

Reference example, not the only acceptable wording: 2, 3 and 4; the stop value is excluded. The body runs three times.

Without opening the old lesson, explain one key idea from Decompose a Goal. Give an example and a mistake that the idea helps you avoid.

Look for: Distinguish a function definition from a function call

Reference example, not the only acceptable wording: Name show_remaining; parameters capacity and used. Arguments 15 and 6 match capacity = 15 and used = 6. Expected printed result 9.

It reads the old 2, adds 3 and stores 5. Assignment **update**s a value; it is not an algebraic equality.

▼ Earlier wording, alternatives and picture transcripts

Teach the learning cycle

Retrieval answer guidance

Trace Tables

Decompose a Goal

Use the student module and workbook section with this exact week title. The worked case, independent question and answers below form one aligned lesson sequence.

What the learner must demonstrate

Trace count=0; while count<4: count=count+2. List values tested by the condition and final count. Would replacing <4 with !=5 terminate?

Concept to explain

Useful repetition needs a clear **update** and **stopping rule**.

Term	Meaning
loop	A structure that repeats instructions according to a count or condition.
update	Change a stored value or state according to a specified operation or new evidence.
stopping rule	A specified condition that ends a repeated process.
infinite loop	A loop that never reaches its stopping condition under the assumed execution model.

Prepare the class

This compact option is not the full-week schedule. Use the main teaching cycle and allow extra sessions for the investigation, independent work and project.

Allow 80 minutes: recall 8; explanation 12; worked demonstration 15; guided practice 20; independent assessment 15; project and reflection 10. Extend the lab into a second session when needed.

Preparation for the connected explanation

Readiness answer

End goal for the pictorial case

This added worked case illustrates the week's main idea. Retain all original and connected topics; schedule its practice within the teaching cycle.

MAIN LESSON · SECTION 2

Teach condition-controlled progress and stopping

> Align the lesson with its while-loop target and keep for-loop review separate.

State the primary while task and scope

Primary while task: test count before each update. For starts 3,4,12 the final outputs are 3,0,0 after zero,one,three updates. The decreasing count eventually reaches ≤3. A !=3 condition from 12 misses three forever because every value is a multiple of four. Use finite paper prefixes for this variant. Earlier range/accumulator material is review, not the main lesson or this week's assessment.

Read the complete script and all condition checks

```
count = 12
while count > 3:
    count = count - 4
print(count)
```

Count before test	count > 3	Next action
12	True	Update to 8
8	True	Update to 4
4	True	Update to 0
0	False	Print 0 and stop

Trace the pictorial version of the same task

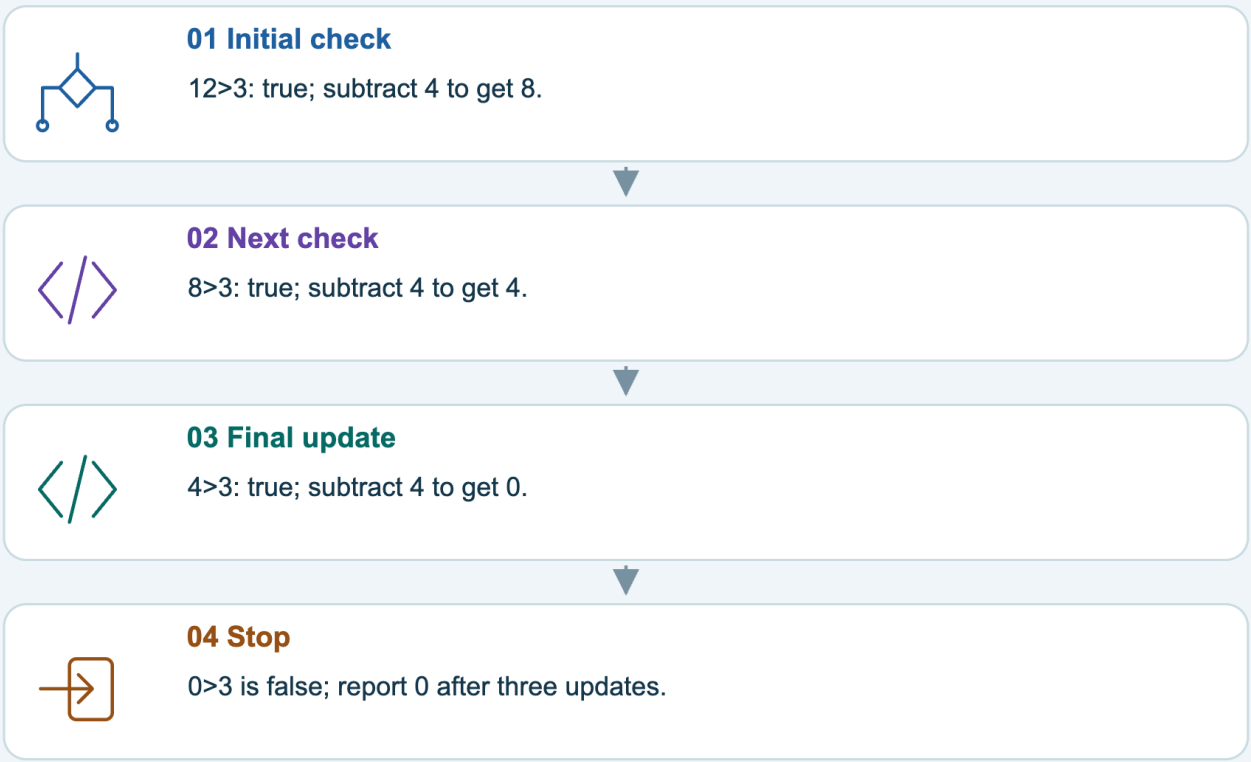
A fictional counter starts at 12 and decreases while above a boundary.

Python: count=12; while count>3: count=count-4; print after the **loop**.

The condition is checked before each subtraction.

AI ACADEMY / AI-2/5 · 02 · TRACE THE CASE

Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Read the picture one step at a time

STEP	WHAT HAPPENS AND WHY
1. Initial check	12>3: true; subtract 4 to get 8.
2. Next check	8>3: true; subtract 4 to get 4.
3. Final update	4>3: true; subtract 4 to get 0.
4. Stop	0>3 is false; report 0 after three updates.

Match each pictured step to the supplied facts. This is a worked teaching case, not an independent assessment.

The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start.

A condition requiring exact equality can be unsafe when repeated **update**s skip that value.

Keep finite-prefix inspection for unbounded defects

Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the defect safely.

Predict the equality-only change

Change: Replace count>3 with count!=3.

Prediction: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

Reason: 12,8,4,0,−4,… are multiples of 4; none equals 3.

Use the diagram to elicit stopping reasoning

Use the diagram and verbal explanation together. Ask learners to match a sentence to a specific visual step; do not assign learners fixed visual or auditory learning styles.

Cover the result. Ask learners to name the **input**s and predict the next step. Uncover one step, connect the icon to the actual procedure, and explain its **evidence**. Then change the supplied condition and compare.

The picture is a representation; it does not document execution of a real AI model.

▼ Earlier wording, alternatives and picture transcripts

Explain the mechanism

Input → **Process** → **Output**

Trace the information. At each stage, ask what changed and how you could test it.

Which stage could fail even when the other two are correct? Explain:

Predict first. Then check each step. A correct answer without visible reasoning is incomplete **evidence**.

2. Repeat while count < 4: show a card; count = count + 1.

3. The **loop** runs 4 times and stops when count becomes 4.

4. If count never changes, the stopping condition never becomes false.

Recalculate, retrace, or restate the reasoning in your own representation:

CHECK Did the **evidence** answer the exact question? Did the total, path, or source record stay consistent?

Concrete teaching case

Start n=0; while n<3, add 1 and record n.

List records and stopping condition.

Exact explanation to model

1,2,3. The next comparison is false. The **update** makes progress toward termination.

Explain the added worked example

Imagine adding numbered counters: first 1, then 2, then 3. A **loop** can repeat the same addition with a different number each time. This week you will predict each change before running a short Python script.

Tracing repeated calculations prepares you to inspect the data **process**ing used in larger AI applications.

In for turn in range(1, 4), Python supplies 1, 2 and 3 in order. The start is included and the stop is excluded. The indented body runs once for each supplied value. The colon and four-space indentation mark that body.

This lesson uses integer range values with the default positive step of 1.

The **output** is 6. Before the first **iteration**, total is 0. Each assignment reads the previous total and adds the current turn. After turn 3, the range is exhausted, so execution continues after the **loop**.

The table records state; it does not create extra instructions.

Key terms

Term	Meaning
Loop	A block of instructions that can run repeatedly
Iteration	One execution of a loop body
Accumulator	A variable that keeps a running result
Initialization	Giving a variable its starting value
Trace table	A record of variable values as instructions execute

Model the reasoning aloud

Use the complete worked example below before asking learners to complete W2. Reveal a small part, invite a **prediction**, then explain the deciding step.

These teacher notes contain solutions; do not reveal the independent case answer during modelling.

Range expression	Values supplied	Number of iterations
range(1, 4)	1, 2, 3	3
range(1, 2)	1	1
range(1, 1)	No values	0
range(4)	0, 1, 2, 3	4

With one argument, range starts at zero. range(4) supplies four values but none is 4. If the worked script uses range(1, 1), the body never runs and the initialized total remains 0.

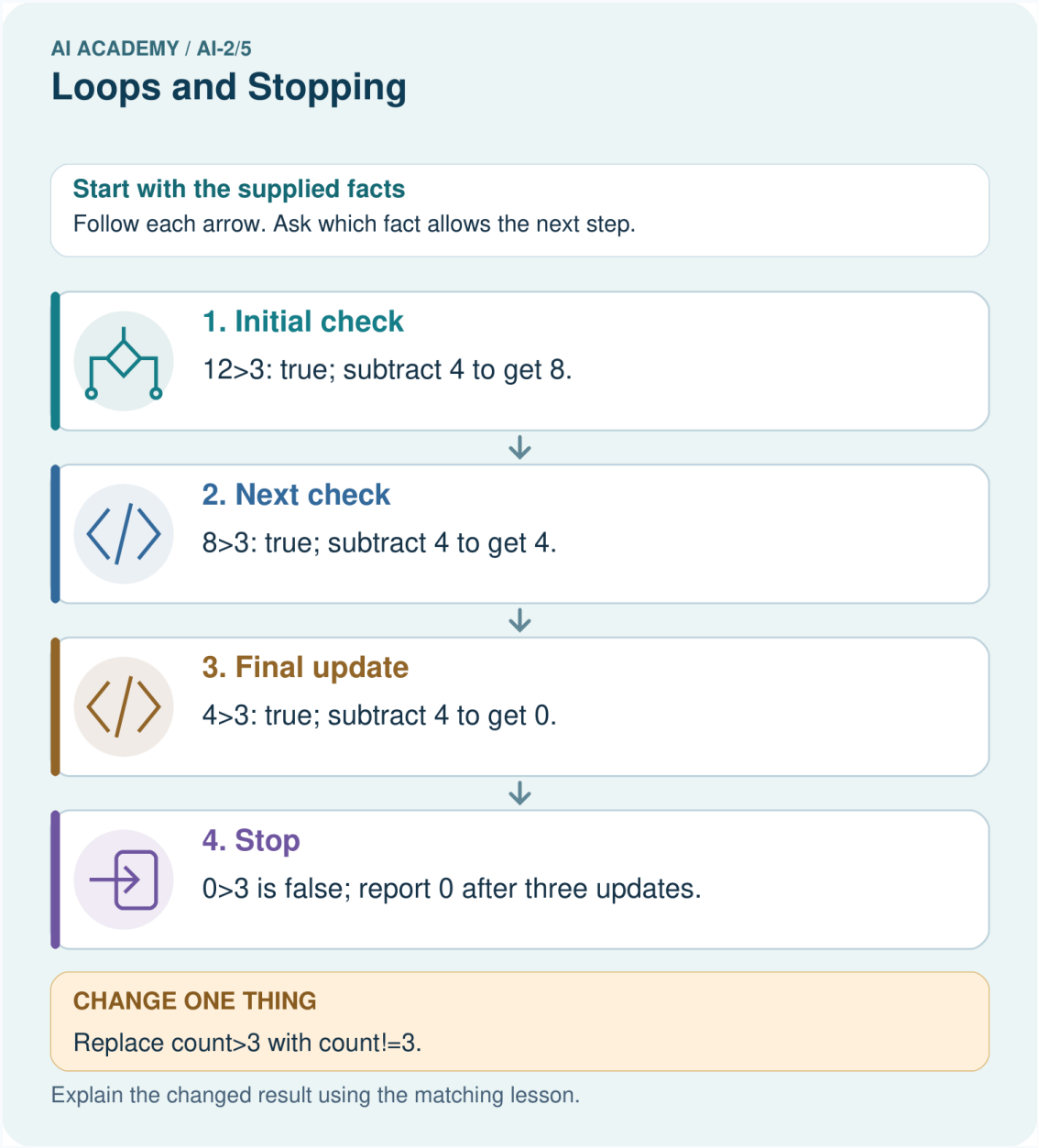
Do not rely on a variable first assigned inside a **loop**: a zero-**iteration** run might never assign it.

> **Replace range(1, 4) with range(1, 3) in the worked script. Predict the supplied values and final total. What stop value is needed to include 3?**

Thinking move	Teacher prompt
Plan	What is given? What is the question? Which procedure fits these facts?
Do	Point to each input and intermediate step in the worked trace. Name the rule that allows the next step.
Check	Does the output answer the question? Which assumption or limitation prevents a larger claim?

A pictorial worked case: Loops and Stopping

The facts we will use



Follow the labelled picture with the matching explanation. Open the vector diagram

Why this result follows

What this example does not establish

Change one thing in the pictured case

Ask while showing the picture

MAIN LESSON · SECTION 3

Guide while traces, then remove help

> Keep earlier for-loop instructions as review and assess the selected increasing-loop case separately.

Completion task and expected reasoning

Using the complete while script, predict fresh starts 3,4,12. Record tested values, truth results, updates and actual **output** separately. Explain the zero-update case.

Required **output**s are 3,0,0 after zero,one,three updates respectively. Record actual runs separately from these expectations.

Start by identifying the given **input**s together. Leave the intermediate steps blank. If needed, model one step, then withdraw that help on the next step. Move to a full trace only when the partial trace is understood.

For partners, alternate explainer and checker; collect a response from every learner before discussion.

Supported diagram check

Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

The original stops as soon as count is not above 3. Exact equality fails because the **update** sequence skips 3.

Changed-case reasoning: From 12, the **loop** never reaches 3 and does not terminate in this toy model. 12,8,4,0,−4,… are multiples of 4; none equals 3.

Collect each learner's explanation before discussion. Use the first missing connection between **input**, deciding step and result to choose support. This question is worked-case practice, not fresh mastery **evidence**.

Collect each learner’s explanation

- Ask: Which facts are given, and which would we have to assume? Learners mark relevant details before discussing answers.

- Say: Predict first. Show the rule, information path or calculation that would produce the result. A correct guess is not yet an explanation.

- In pairs, one learner solves while the other checks **evidence**; exchange roles. Remove the worked answer before independent assessment.

▼ Earlier wording, alternatives and picture transcripts

Demonstrate and guide practice

Board sequence

Write the given facts: Start n=0; while n<3, add 1 and record n.

Write the question to explain: List records and stopping condition.

Reveal after **prediction**s: 1,2,3. The next comparison is false. The **update** makes progress toward termination.

Use the earlier changed question as optional later practice. Reserve Changed independent case for the first independent attempt before hints or feedback.

Classroom dialogue

Reveal reasoning one step at a time. Explain why a tempting alternative fails. Write units and denominators for arithmetic, trace conditions for decisions, and identify supporting sources for claims.

Model these guided questions

range(1, 3) supplies 1 and 2, giving total 3. Stop 4 includes 3. Exit: the reset bug agrees for one **iteration** because there are no earlier additions to lose. range(1, 4) should give 6; the faulty version gives 3.

Case-specific teaching sequence

Stage	Teacher action
Session 1 start to 7 minutes	Retrieve assignment using the prerequisite.
7 to 20 minutes	Model range with numbered cards and complete W1.
20 to 33 minutes	Trace each row of the worked addition and predict output.
33 to 45 minutes	Complete W2 and compare predicted and observed values.
Session 2 start to 8 minutes	Retrieve included start and excluded stop.
8 to 20 minutes	Trace the reset bug, complete W3 and discuss the guided response.
20 to 35 minutes	Collect independent W4 code and zero one and several iteration traces.
35 to 45 minutes	Use W5 and the exit to examine assumptions and revealing tests.

Reduce help deliberately

Individual reasoning check

Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported. Point to the step or supplied fact that decides; explain your repair if needed.

Expected correction: List the supplied values before tracing; range(1, 4) excludes 4.

Collect explanations, not only yes/no votes. If a misunderstanding is common, re-model the relevant step for the class.

If it affects a few learners, give a short targeted group explanation while others solve a more demanding variation.

Pictorial practice question

Reasoned teacher answer

Run the supplied investigation

> Keep expected results, observed paper traces and remaining limits distinguishable.

Supplied materials and answer guidance

Correct decreasing-count script: starts 3,4,12 produce final 3,0,0 with zero,one,three subtractions. The paper incrementing-card task tests 0,1,2,3,4 and displays cards at 0,1,2,3 before updates. Without an update, initially true conditions do not become false. For the !=3 variant, trace only a finite prefix; multiples of four cannot equal three. Keep actual runs and old versions. The selected independent case is the increasing count from 1 by 3 toward <8; reserve it for an uncoached attempt.

Run and record

1. Predict → Test → Record → Explain
2. Keep expected and actual results separate. Surprises are useful data when recorded honestly.
3. Walk a robot around a four-card **loop**. Record the **update** each time; remove the **update** to observe the failure; restore it and retest.
4. What did the hardest test teach you that the normal test did not?

Keep an inspectable record

Record	What to capture
Input or evidence	Exact data, instruction or claim
Expected result	Answer or safe fallback declared before the run
Observed result	Actual result, including errors

Record	What to capture
Explanation	Rule, calculation or evidence explaining the outcome
Next test	One justified change and a new prediction

Diagnose the first misunderstanding

The stop value is included: List the supplied values before tracing; range(1, 4) excludes 4.

Initialization belongs inside every **iteration**: Resetting the **accumulator** loses earlier work.

An unindented print repeats with the body: Indentation determines which statements repeat.

Zero **iteration**s mean the entire program does nothing: **Initialization** and statements after the **loop** still execute.

Choose support from the response

What the response shows	Next teaching action
Missing or misread facts	Read the case aloud, mark supplied versus unknown facts, and let the learner restate it.
Wrong procedure	Return to the deciding step in the worked case. Contrast it with the proposed incorrect explanation.
Arithmetic or implementation error	Trace intermediate values or lines; preserve the first mismatch and retry that step.
Correct result without a reason	Ask for a trace and a changed case before marking independent understanding.

What the response shows	Next teaching action
An unsupported wider claim	Ask which supplied evidence supports the claim and which further observation would be needed.

Offer an oral explanation, drawing, enlarged visual or fewer examples when that removes a reading/writing barrier. Keep the same reasoning goal.

Stretch secure learners by changing a boundary or assumption and asking them to defend the effect, not simply by assigning more of the same questions.

▼ Earlier wording, alternatives and picture transcripts

Run the weekly evidence lab

Words You Can Use to Reason

A definition matters when you can recognize the idea, use it, and contrast it with a near-miss.

Make the lab runnable

Use the supplied investigation for shared practice. Keep predictions and observed paper-trace results separate. Reserve Changed independent case for the individual first attempt before feedback; do not use the reserved case as a paired lab task.

Diagnose and reteach

For guessing, return to given facts and request a trace. For vocabulary without application, use one example and one non-example. For arithmetic errors, check the setup before the calculation.

Finish with a different uncoached case before recording mastery.

Address these additional misconceptions

Match the support to the first error

Assess the selected new case

> Keep this key until after the first attempt; accept valid alternatives with defensible reasoning.

Selected independent question and key

Selected case: Start count at 1. While count < 8, add 3. Trace the condition and values. What happens if the condition becomes count != 8?

The condition is true at 1,4,7, then false at 10. Final count is 10. With !=8 the sequence never reaches 8: all values are 1 modulo 3, so this **loop** does not terminate.

Judge explanation, trace and limits

Dimension	Evidence required
Explanation	Names the applicable idea or procedure and ties it to the supplied facts.
Trace and result	Preserves the given inputs and shows the relevant intermediate steps; distinguishes a prediction from an observed execution.
Limits	States the assumptions, missing evidence or conditions under which the result would change.

Record each dimension as not yet, with support, or independent. Accept a valid alternative procedure with a defensible trace.

Do not penalise an honestly recorded model failure if the learner correctly explains its **evidence** and limits.

Give one usable repair action

Name the earliest unsupported step and give one action to repair it. Have the learner make that repair now and explain why. Then assess the relevant skill on a fresh, fully specified case.

Confirm it was not previously practised with feedback; solve and check the new case before assigning it. A follow-up design prompt is a starting brief, not automatically an equivalent test.

W1 Explain the range

For count=12, while count>3 subtract 4, list every value tested by the condition, including the final false test. How many updates occur, and what is printed?

Test 12,8,4,0; three updates; print 0. The false check at 0 prevents another subtraction.

W2 Trace before running

Using the complete while script, predict fresh starts 3,4,12. Record tested values, truth results, updates and actual **output** separately. Explain the zero-update case.

Required **output**s are 3,0,0 after zero,one,three updates respectively. Record actual runs separately from these expectations.

W3 Repair the reset

Replace count>3 with count!=3, starting at 12 and subtracting 4. Trace only five updates on paper, explain why equality is missed and restore the appropriate condition.

The prefix is 12,8,4,0,-4,-8. Values remain multiples of four and never equal three. Restore >3; keep the faulty prefix and label the repaired run.

W5 Check a contract limitation

A loop with no update is tested only with count=3 and stops. Does that establish termination from count=12? Explain and choose starts giving zero, one and several updates in the correct script.

No. At 3 the initial condition is false, hiding the missing update. From 12 it stays true without progress. Correct starts 3,4,12 exercise zero,one,three updates.

▼ Earlier wording, alternatives and picture transcripts

Assess and explain the answers

Worked case answer

1,2,3. The next comparison is false. The **update** makes progress toward termination.

Additional transfer practice question

Start at count 0. While count is less than 3, add 1 and display count. List the displays and the stopping value.

Reasoned practice answer

Displays are 1, 2 and 3. The next condition check finds 3 is not less than 3, so the **loop** stops.

What counts as evidence

Criterion	0 points	1 point	2 points
Result	Incorrect or absent	Partly correct	Correct with necessary conditions
Reasoning	No supporting trace	Some steps	Clear mechanism, calculation or source support
Limit	Overclaims certainty	Vague limitation	Relevant uncertainty or missing fact
Transfer	Copies without adapting	Adapts with prompts	Solves a changed case independently

Use 6/8 as a classroom checkpoint, with 2 required for both Result and Reasoning. This is a proposed teaching rule, not a validated certification threshold.

Do not count a coached answer as independent mastery; correct unsafe or unsupported claims regardless of total.

Feedback

Ask the learner to find the **evidence** that challenges this claim: “A less-than-3 **loop** can never display 3.” Keep the first attempt and an uncoached retry separate.

Original transfer answer

The condition tests 0,2,4; the body runs twice and ends at 4. With !=5, even counts never equal 5, so it does not terminate under unbounded arithmetic.

A **stopping rule** should match the **update** and include resource bounds in an actual application.

Explanations for the additional workbook examples

W1 Explain the range

W2 Trace before running

W3 Repair the reset

W4 Build a repeated addition independently

A box starts with 5 counters. Each round adds 2 counters. Use rounds = 3, an initialized total and for turn in range(rounds). Print the final total once after the **loop**.

Trace the totals and predict fresh runs with rounds 0, 1 and 3. Assume rounds is a nonnegative integer.

Initialize total = 5 before the **loop**; add 2 inside it. For three rounds, turn values 0, 1, 2 give totals 7, 9, 11. Final **output**s for 0, 1, 3 rounds are 5, 7, 11. Print is unindented.

W5 Check a contract limitation

Assess independent transfer

Selected case answer

Feedback that the learner uses

MAIN LESSON · SECTION 6

Connect, adapt and return

> Retain the project evidence and revisit the reasoning on a changed case.

Project evidence to keep

Loop through every candidate book, **update** its score, then stop at the end of the list.

Keep the decision, **input**, result, explanation and remaining limitation in the project **evidence** record. Retrieve this record next week rather than restarting the project.

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s.

Support while preserving the reasoning goal

Trace one record or condition at a time with counters and labelled columns. Stretch: introduce a tie, boundary or missing field and declare its handling policy before rerunning.

Use counters and let the learner move a pointer through one instruction at a time. Offer a blank trace grid and permit oral explanation.

Extension: put print inside the correct **loop** and explain why zero rounds produces no **output** while an outside print still does.

The selected independent case tests 1,4,7,10 and stops at 10 under <8. Under !=8 it never reaches eight because all states are 1 modulo 3. Assess a fresh finite prefix and a stopping argument, recording support; do not use the old repeated-addition exercise as this selected assessment.

Plan the delayed check

Suggested return points in this level: ai-2/6, ai-2/8, ai-2/11. Use actual classroom dates, adjusting for holidays and gaps. One, three and six teaching weeks are scheduling choices, not a claimed optimal spacing.

At a delayed check, ask for the idea from memory and a changed application. Compare with the earlier independent response and record support used.

For a new level, revisit the previous capstone before checking the new prerequisites.

Record	Teacher evidence
Learner code and date	
Concept and case used; prior exposure	

Record	Teacher evidence
Explanation / trace / limits	
Support and first repair	
Changed case and delayed outcome	
Next teaching action	

Use the **evidence** to choose reteaching or the next step. A completed page, confidence rating or attractive project does not by itself demonstrate understanding.

Bridge to next week

Ask for one decision the learner can now justify and one situation requiring more **evidence** or human review. Confidence ratings are separate from correct independent performance.

Next: Week 6: Debugging Strategies. Retrieve today's **evidence** and identify what carries forward and what is new.

Week 6 develops debugging and revealing function tests; Week 7 develops structured records and schemas.

▼ Earlier wording, alternatives and picture transcripts

Connect and extend learning

Transparent Book Recommender

Support and stretch

Exit reflection

Bridge to the next week

Support and fresh reassessment

Plan the delayed return

Project evidence from this case

Reference bank and extra practice

Choose the reference that helps your next step.

▼ **Reference: alternative planning checklist**

REFERENCE · SECTION 7

Reference: alternative planning checklist

> Use as a reminder alongside the main route rather than teaching the cycle twice.

Preparation and schedule

Teaching target: Trace variable values before and after each **loop iteration**

Prepare the labelled worked-case picture, the matching workbook, fictional **input** cards and a place for individual responses. No paid AI account is required for the paper reasoning activities.

Session · Plan · Resources

> **Session:** Session 1 · Understand · 45 min
>
> **Plan:** 0–7: recall and prerequisite; 7–15: inspect the concrete case and predict; 15–30: teacher models the worked trace; 30–40: learners explain a step using the text and visual; 40–45: check the core idea.
>
> **Resources:** Use Your mission, Notice the evidence, See how it works, Compare the cases, Words that unlock the idea and Follow a worked example; record the brief recall in the workbook.

> **Session:** Session 2 · Practise · 45 min

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> **Resources:** Use Run the investigation, Connect your project, Show what you know, Study a complete case and Try a new case independently. Allow extra time for the full task set. Use a coding lab only when this week actually provides one; Your lesson route is the planning section.

Practice and assessment reminders

Before class, solve the supported practice: Copy the worked script. Change the range to range(2, 5). Record turn, total before and total after each **iteration**. Predict the single **output**, then run and compare.

Reasoned practice answer: Rows are 2 / 0 / 2, 3 / 2 / 5, 4 / 5 / 9. Final **output** 9. Keep the observed result distinct from the **prediction**.

Check every learner before discussion: Use the worked case for **Loop**s and **trace table**s. Proposed explanation: “The stop value is included”. Decide whether this explanation is supported.

Point to the step or supplied fact that decides; explain your repair if needed.

Misconception repair: List the supplied values before tracing; range(1, 4) excludes 4.

Support: read the exact case aloud, enlarge the labelled picture, supply **input** cards and model only the first deciding step. Accept a spoken explanation or drawing; retain the reasoning goal.

Extension: Replace count>3 with count!=3. Require a **prediction** and a reason before comparing. Expected result: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

Reason: 12,8,4,0,−4,… are multiples of 4; none equals 3.

Independent assessment: withhold the reserved answer until the learner has attempted the new case. Record support separately from correctness.

Criterion	Not yet	With support	Independent
Explanation	Names an answer without a deciding fact	Uses a hint to link fact and decision	Explains the deciding fact in own words
Trace or test	Cannot connect input and result	Completes steps with prompts	Traces or tests a changed case accurately
Limits and revision	Claims more than evidence supports	Names a limit after a prompt	States a defensible limit and useful next test

Project checkpoint: Frame the need. Name the user, the problem, allowed **input**s and a safe simple starting method.

Use the weekly teacher answer for topic-specific reasoning. A saved progress tick is an activity record; judge mastery from the learner's explanation and **evidence**.

▼ **Reference: feedback and records**

REFERENCE · SECTION 8

Reference: feedback and records

> Use digital tools where they help retain an explanation and its evidence.

Keep practice and independent evidence separate

Use the two formative questions as supported practice, not as independent mastery scores. Discuss the deciding **evidence** rather than accepting a guessed selection.

The feedback comes from the reviewed public worked case. Do not reveal the separately reserved independent teacher answer until the learner has attempted that case.

Repair and portfolio records

Use the earlier worked-explanation links to address a specific missing building block, then return to the current question. Adapt this suggestion after observing the learner; it is not an automatic diagnosis.

Collect the completed-work portfolio as **evidence**. Record support separately from correctness, and assess explanation, trace/test and limitations. Progress ticks and quiz selections alone do not establish mastery.

<details>
<summary>Earlier wording, alternatives and picture transcripts</summary>

Ask students to predict before running code. Preserve errors and actual **output**s; collect a changed test and an explanation. The branch-boundary and indexing lessons deliberately include failure cases.
</details>

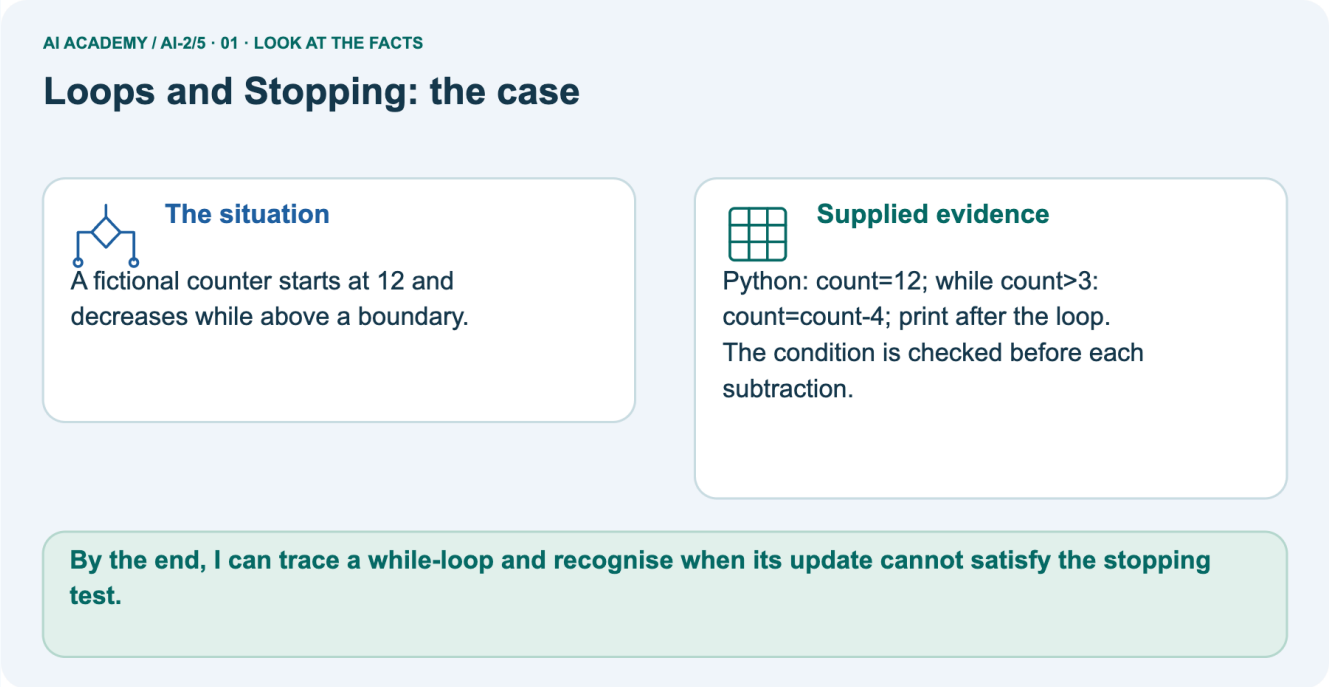
<details>
<summary>Reference: teach with the six pictures</summary>
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REFERENCE · SECTION 9

Reference: teach with the six pictures

> Choose the picture for the teaching move and connect it to the explanation.

01. Loops and Stopping: the case



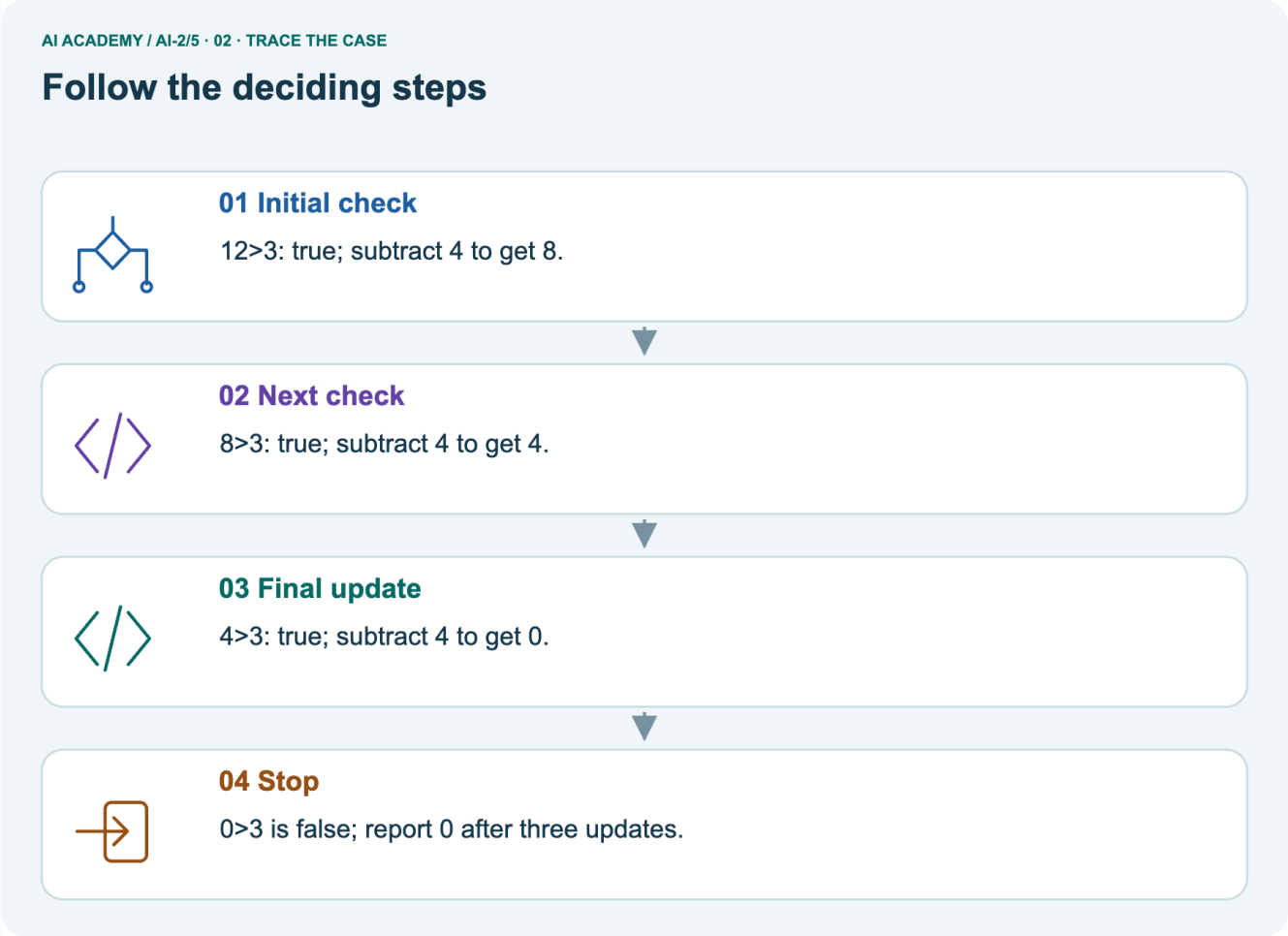
Follow the labelled picture with the matching explanation. Open the vector diagram

A fictional counter starts at 12 and decreases while above a boundary. Supplied facts: Python: count=12; while count>3: count=count-4; print after the **loop**. The condition is checked before each subtraction.

Point to the given fact that matters most. Explain why it matters.

Teacher prompt: Point to the given fact that matters most. Explain why it matters. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

02. Follow the deciding steps



Follow the labelled picture with the matching explanation. Open the vector diagram

Initial check: 12>3: true; subtract 4 to get 8. → Next check: 8>3: true; subtract 4 to get 4. → Final **update**: 4>3: true; subtract 4 to get 0. → Stop: 0>3 is false; report 0 after three **update**s.

Point to each numbered stage and explain which recorded **input** it uses.

Teacher prompt: Point to each numbered stage and explain which recorded **input** it uses. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

03. State after each addition

AI ACADEMY / AI-2/5 · 03 · INSPECT THE LESSON MODEL

State after each addition

turn	total before	total after
1	0	1
2	1	3
3	3	6

Initialize total to 0 before the loop. The unindented print runs once afterward.

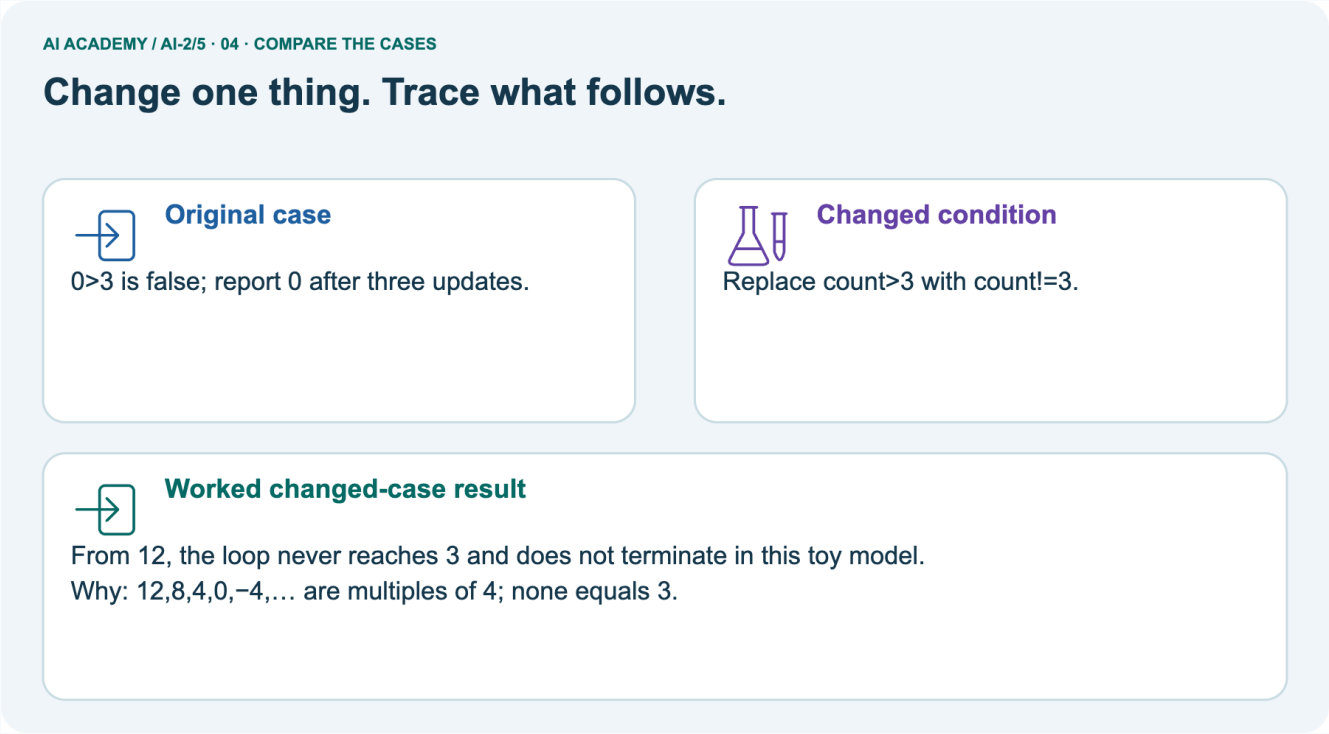
Follow the labelled picture with the matching explanation. Open the vector diagram

State after each addition. 1 | 0 | 1. 2 | 1 | 3. 3 | 3 | 6

Read the labels and values. Explain the relationship the model shows.

Teacher prompt: Read the labels and values. Explain the relationship the model shows. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

04. Change one thing. Trace what follows.



Follow the labelled picture with the matching explanation. Open the vector diagram

Original case: 0>3 is false; report 0 after three **update**s. Changed condition: Replace count>3 with count!=3. Expected result: From 12, the **loop** never reaches 3 and does not terminate in this toy model.

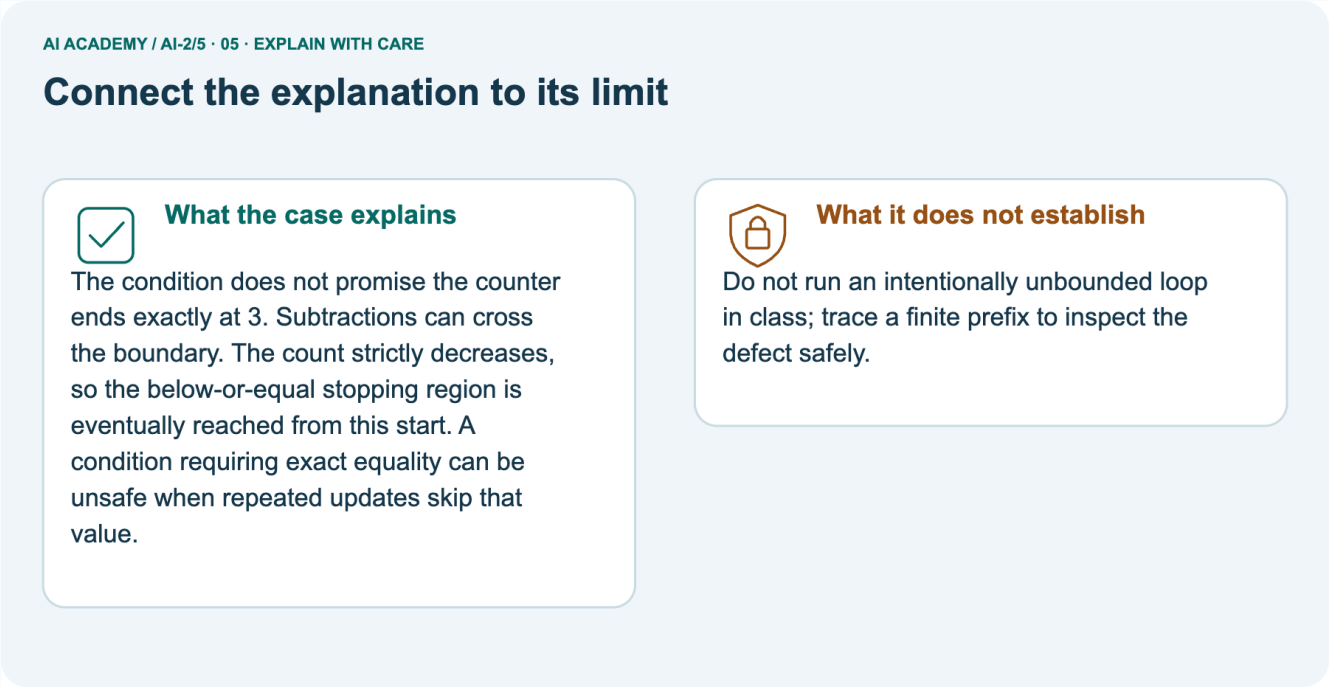
Reason: 12,8,4,0,−4,… are multiples of 4; none equals 3.

Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude.

Teacher prompt: Predict before checking the worked changed-case result. Name what changed and what the supplied facts allow you to conclude. Ask learners to describe one label or arrow in ordinary words.

Use the adjacent transcript if the image is difficult to see. Changed-case reasoning: 12,8,4,0,−4,… are multiples of 4; none equals 3.

05. Connect the explanation to its limit



Follow the labelled picture with the matching explanation. Open the vector diagram

The condition does not promise the counter ends exactly at 3. Subtractions can cross the boundary. The count strictly decreases, so the below-or-equal stopping region is eventually reached from this start.

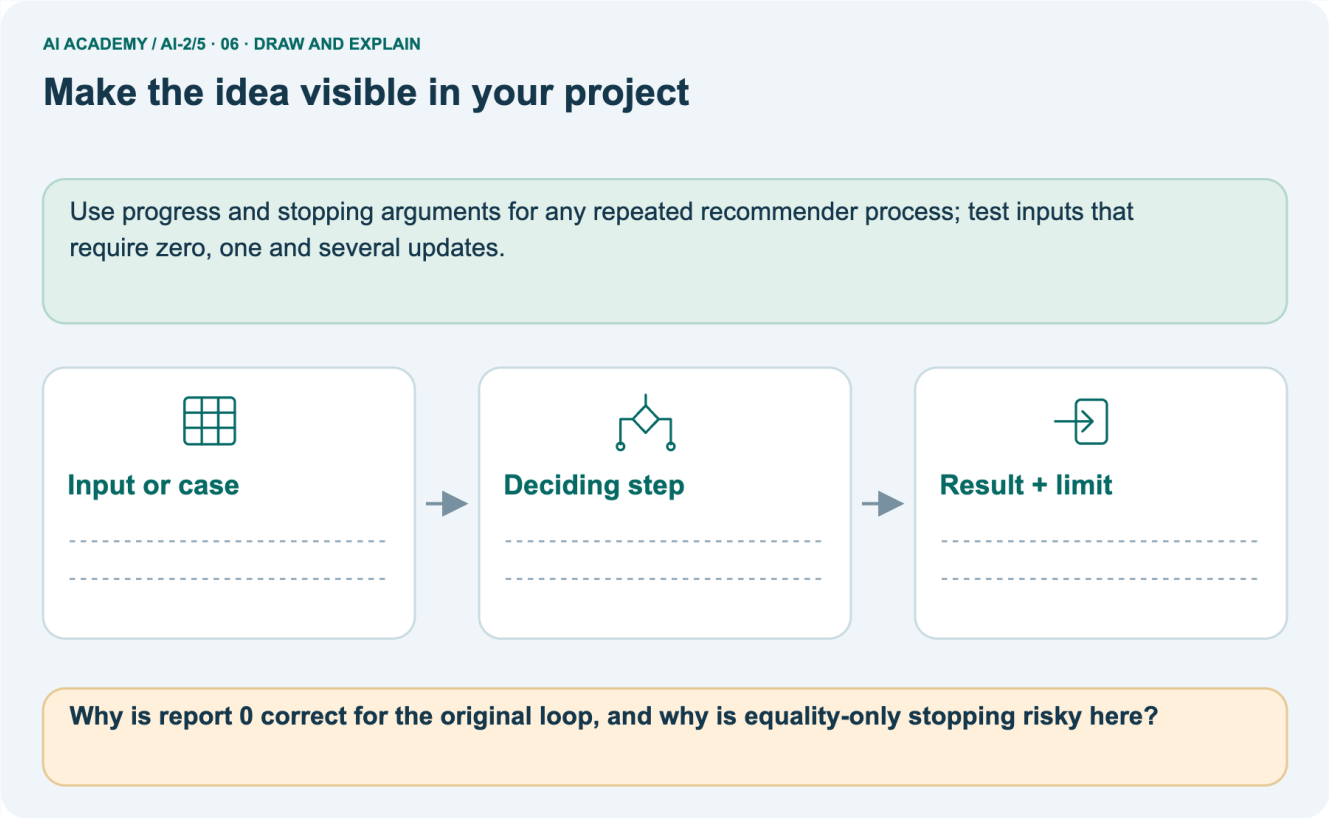
A condition requiring exact equality can be unsafe when repeated **update**s skip that value. Limit: Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the defect safely.

Explain the case, then point to a limit on the claim.

Teacher prompt: Explain the case, then point to a limit on the claim. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Keep the stated scope: Do not run an intentionally unbounded **loop** in class; trace a finite prefix to inspect the defect safely.

06. Make the idea visible in your project



Follow the labelled picture with the matching explanation. Open the vector diagram

Use progress and stopping arguments for any repeated recommender **process**; test **input**s that require zero, one and several **update**s. Draw your **input**, deciding step, result and limit.

Supported question: Why is report 0 correct for the original **loop**, and why is equality-only stopping risky here?

Draw your project trace and justify the deciding step using **evidence**.

Teacher prompt: Draw your project trace and justify the deciding step using **evidence**. Ask learners to describe one label or arrow in ordinary words. Use the adjacent transcript if the image is difficult to see.

Supported practice key: The original stops as soon as count is not above 3. Exact equality fails because the **update** sequence skips 3.

▼ Earlier wording, alternatives and picture transcripts

Use one picture at a time. Hide a worked result before asking for a **prediction**, then reveal it for feedback. These are supported teaching cases; retain the separate independent assessment for a fresh first attempt.

01. Loops and Stopping: the case

02. Follow the deciding steps

03. State after each addition

04. Change one thing. Trace what follows.

05. Connect the explanation to its limit

06. Make the idea visible in your project

▼ Optional interactive practice and completed work

Save or print your responses for your portfolio. Progress and drafts on the hosted website stay in the current browser. Use only fictional or teacher-approved data.